

CHAPTER 9

Organic Chemistry–Some Basic Principles and Techniques

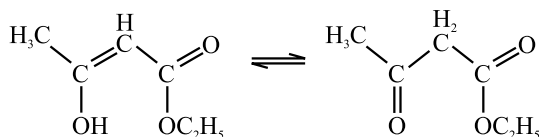
Tetravalence of Carbon (Shapes of Organic Compounds)

1. Match List I with List II. (2024)

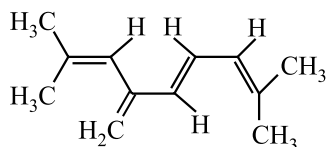
List I (Molecule)		List II (Number and types of bonds between two carbon atoms)	
A.	ethane	I.	one σ -bond and two π -bonds
B.	ethene	II.	two π -bonds
C.	carbon molecule, C_2	III.	one σ -bond
D.	ethyne	IV.	one σ -bond and one π -bond

Choose the correct answer from the options given below:

- a. A-III, B-IV, C-II, D-I b. A-III, B-IV, C-I, D-II
c. A-I, B-IV, C-II, D-III d. A-IV, B-III, C-II, D-I
2. The number of σ bonds, π bonds and lone pair of electrons in pyridine, respectively are (2023)
a. 12, 2, 1 b. 11, 2, 0 c. 12, 3, 0 d. 11, 3, 1
3. Which of the following molecules represents the order of hybridisation sp^2 , sp^2 , sp , sp from left to right atoms? (2018)
a. $HC \equiv C - C \equiv CH$ b. $CH_2 = CH - C \equiv CH$
c. $CH_3 - CH = CH - CH_3$ d. $CH_2 = CH - CH = CH_2$
4. Which of the following species contains equal number of σ -bonds and π -bonds? (2015)
a. $(CN)_2$ b. $CH_2(CN)_2$ c. HCO_3^- d. XeO_4
5. The enolic form of ethyl acetoacetate as shown below has (2015)



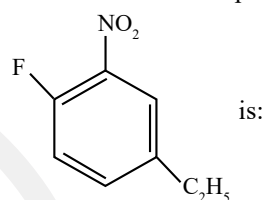
- a. 9 σ -bonds and 2 π -bonds b. 9 σ -bonds and 1 π -bonds
c. 18 σ -bonds and 2 π -bonds d. 16 σ -bonds and 1 π -bond
6. The total number of π -bond electrons in the following structure is (2015)



- a. 12 b. 16 c. 4 d. 8

Classification, Nomenclature of Organic Compounds

7. The correct IUPAC name of the compound (2024 Re)

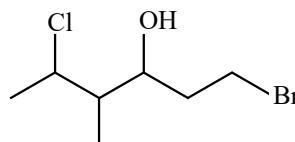


is:

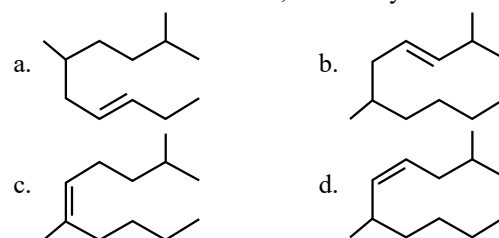
- a. 4-ethyl-1-fluoro-2-nitrobenzene
b. 4-ethyl-1-fluoro-6-nitrobenzene
c. 3-ethyl-6-fluoro-1-nitrobenzene
d. 1-ethyl-4-fluoro-3-nitrobenzene
8. A compound with a molecular formula of C_6H_{14} has two tertiary carbons. Its IUPAC name is: (2024)
a. 2,3-dimethylbutane b. 2,2-dimethylbutane
c. n-hexane d. 2-methylpentane
9. What is the hybridization shown by C_1 and C_2 carbons, respectively in the given compound? (2022 Re)

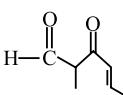


- a. sp^3 and sp^3 b. sp^2 and sp^3
c. sp^2 and sp^2 d. sp^3 and sp^2
10. The correct IUPAC name of the following compound is: (2022)

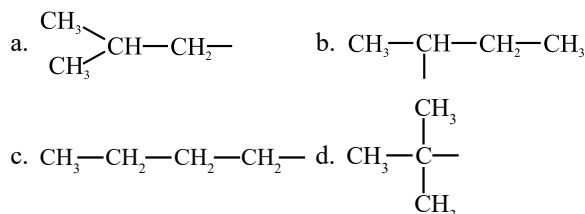


- a. 6-bromo-4-methyl-2-chlorohexan-4-ol
b. 1-bromo-5-chloro-4-methylhexan-3-ol
c. 6-bromo-2-chloro-4-methylhexan-4-ol
d. 1-bromo-4-methyl-5-chlorohexan-3-ol
11. The correct structure of 2,6-Dimethyl-dec-4-ene is: (2021)

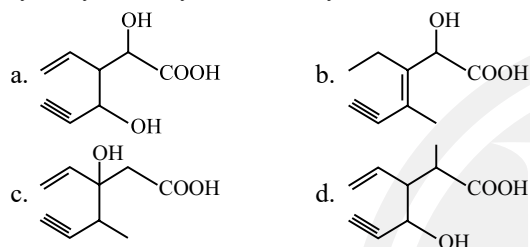


12. The IUPAC name of the compound  is
 a. 3-keto-2-methylhex-5-enal
 b. 3-keto-2-methylhex-4-enal
 c. 5-formylhex-2-en-3-one
 d. 5-methyl-4-oxohex-2-en-5-al

13. The structure of isobutyl group in an organic compound is: (2013)

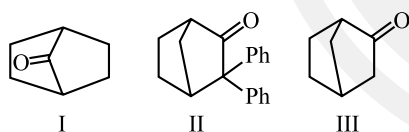


14. Structure of the compound whose IUPAC name is 3-Ethyl-2-hydroxy-4-methylhex-3-en-5-ynoic acid is: (2013)



Isomerism

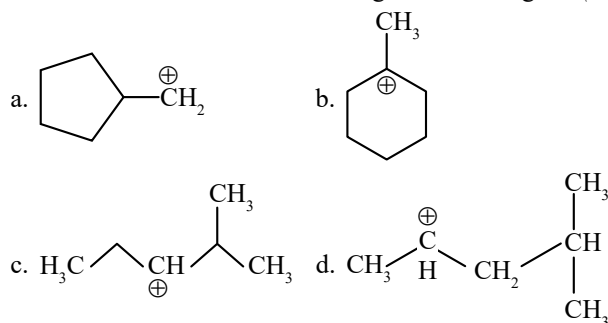
15. The compound which shows metamerism is: (2021)
 a. C_3H_8O b. C_3H_6O c. $C_4H_{10}O$ d. C_5H_{12}
 16. Which among the given molecules can exhibit tautomerism? (2016 - II)



- a. III only b. Both I and III
 c. Both I and II d. Both II and III

Fundamental Concepts of Organic Reactions

17. Methyl group attached to a positively charged carbon atom stabilizes the carbocation due to (2024 Re)
 a. -I inductive effect b. electromeric effect
 c. hyperconjugation d. mesomeric effect
 18. The most stable carbocation among the following is: (2024)



19. Given below are two statements: (2023-Manipur)

Statement-I: In a organic compound, when inductive and electromeric effects operate in opposite directions, the inductive effect predominates.

Statement-II: Hyperconjugation is observed in o-xylene.

In the light of the above statements, choose the correct answer from the options given below:

- a. **Statement-I** is true but **Statement-II** is false.
 b. **Statement-I** is false but **Statement-II** is true.
 c. Both **Statement-I** and **Statement-II** are true.
 d. Both **Statement-I** and **Statement-II** are false.

20. Given below are two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R). (2022 Re)

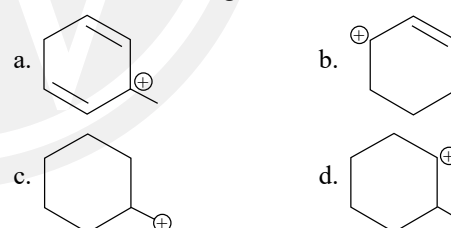
Assertion (A): Chlorine is an electron withdrawing group but it is ortho, para directing in electrophilic aromatic substitution.

Reason (R): Inductive effect of chlorine destabilises the intermediate carbocation formed during the electrophilic substitution, however due to the more pronounced resonance effect, the halogen stabilises the carbocation at ortho and para positions.

In the light of the above statements, choose the most appropriate answer from the options given below :

- a. (A) is not correct but (R) is correct.
 b. Both (A) and (R) are correct and (R) is the correct explanation of (A).
 c. Both (A) and (R) are correct but (R) is not the correct explanation of (A).
 d. (A) is correct but (R) is not correct.

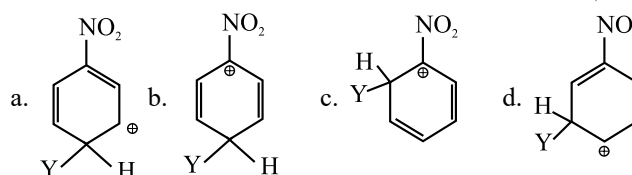
21. Which of the following is the most stable carbocation? (2022 Re)



22. A tertiary butyl carbocation is more stable than a secondary butyl carbocation because of which of the following? (2020)

- a. +R effect of $-CH_3$ groups b. -R effect of $-CH_3$ groups
 c. Hyperconjugation d. -I effect of $-CH_3$ groups

23. Which of the following carbocations is expected to be most stable? (2018)



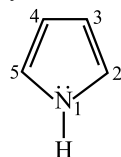
24. Which of the following is correct with respect to -I effect of the substituents? (R = alkyl) (2018)

- a. $-NH_2 < -OR < -F$ b. $-NR_2 < -OR < -F$
 c. $-NR_2 > -OR > -F$ d. $-NH_2 > -OR > -F$

25. The correct statement regarding electrophile is: (2017-Delhi)
- Electrophile can be either neutral or positively charged species and can form a bond by accepting a pair of electrons from a nucleophile
 - Electrophile is a negatively charged species and can form a bond by accepting a pair of electrons from a nucleophile
 - Electrophile is a negatively charged species and can form a bond by accepting a pair of electrons from another electrophile
 - Electrophiles are generally neutral species and can form a bond by accepting a pair of electrons from a nucleophile

26. In pyrrole, the electron density is maximum on (2016 - II)

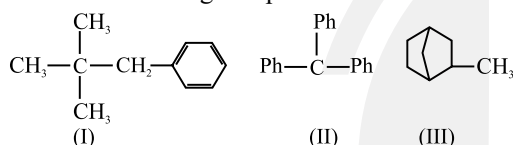
- 2 and 3
- 3 and 4
- 2 and 4
- 2 and 5



27. Which of the following statements is not correct for a nucleophile? (2015 Re)

- Nucleophiles are not electron seeking
- Nucleophile is a Lewis acid
- Ammonia is a nucleophile
- Nucleophiles attack low electron density sites

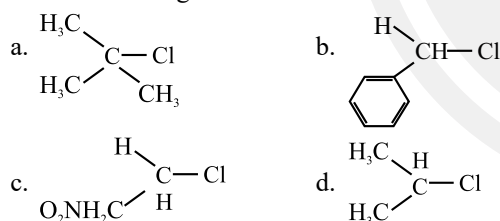
28. Consider the following compounds



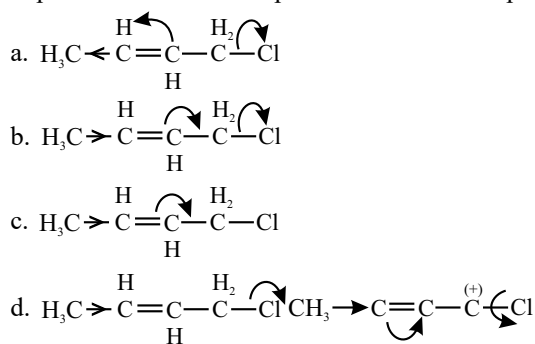
Hyperconjugation occurs in: (2015)

- II only
- III only
- I and III
- I only

29. In which of the following compounds, the C - Cl bond ionization shall give most stable carbonium ion? (2015)



30. Which of the following is the most correct electron displacement for a nucleophilic reaction to take place? (2015)

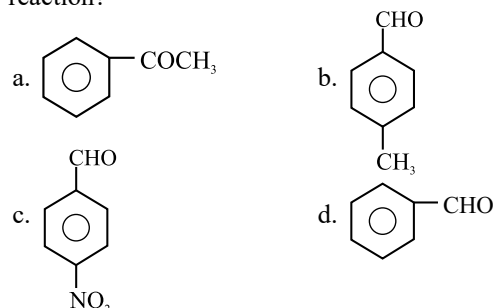


31. Treatment of cyclopentanone with methyl lithium

gives which of the following species? (2015)

- Cyclopentanonyl radical
- Cyclopentanonyl biradical
- Cyclopentanonyl anion
- Cyclopentanonyl cation

32. Which one is most reactive towards nucleophilic addition reaction? (2014)



33. Some *meta*-directing substituents in aromatic substitution are given. Which one is most deactivating? (2013)

- COOH
- NO₂
- C≡N
- SO₃H

Methods of Purification, Qualitative and Quantitative Analysis of Organic Compounds

34. A steam volatile organic compound which is immiscible with water has a boiling point of 250°C. During steam distillation, a mixture of this organic compound and water will boil: (2024 Re)

- above 100°C but below 250°C
- above 250°C
- at 250°C
- close to but below 100°C

35. On complete combustion, 0.3 g of an organic compound gave 0.2 g of CO₂ and 0.1 g of H₂O. The percentage composition of carbon and hydrogen in the compound, respectively is: (2024 Re)

- 4.07% and 15.02%
- 18.18% and 3.70%
- 15.02% and 4.07%
- 3.70% and 18.18%

36. On heating some solid substances change from solid to vapour state without passing through liquid state. The technique used for the purification of such solid substances based on the above principle is known as (2024)

- Distillation
- Chromatography
- Crystallization
- Sublimation

37. In Lassaigne's extract of an organic compound, both nitrogen and sulphur are present, which gives blood red colour with Fe³⁺ due to the formation of (2023)

- [Fe(SCN)]²⁺
- Fe₄[Fe(CN)₆]₃.xH₂O
- NaSCN
- [Fe(CN)₅NOS]⁴⁻

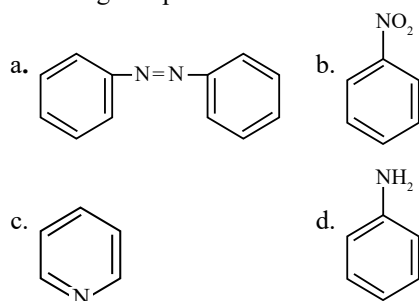
38. Match List-I with List-II (2023-Manipur)

	List-I (Mixtures/Sample)		List-II (Technique used for purification)
A.	Glycerol from spent lye	I.	Steam distillation
B.	Chloroform + Aniline	II.	Fractional distillation
C.	Fraction of crude oil	III.	Distillation under reduced pressure
D.	Aniline + Water	IV.	Distillation

Choose the **correct** answer from the options given below:

- A-III, B-IV, C-II, D-I
- A-IV, B-II, C-I, D-III
- A-I, B-II, C-III, D-IV
- A-I, B-III, C-II, D-IV

39. The Kjeldahl's method for the estimation of nitrogen **can be used** to estimate the amount of nitrogen in which one of the following compounds? (2022)



40. Paper chromatography is an example of: (2020)
- Partition chromatography
 - Thin layer chromatography
 - Column chromatography
 - Adsorption chromatography
41. A liquid compound (x) can be purified by steam distillation only if it is (2020-Covid)

- Not steam volatile, miscible with water
- Steam volatile, miscible with water
- Not steam volatile, immiscible with water
- Steam volatile, immiscible with water

42. The most suitable method of separation of 1 : 1 mixture of ortho and para-nitrophenols is (2017-Delhi)
- Steam distillation
 - Sublimation
 - Chromatography
 - Crystallisation
43. In Duma's method for estimation of nitrogen, 0.25 g of an organic compound gave 40 mL of nitrogen collected at 300 K temperature and 725 mm pressure. If the aqueous tension at 300 K is 25 mm, the percentage of nitrogen in the compound is: (2015)
- 18.20
 - 16.76
 - 15.76
 - 17.36
44. In the Kjeldahl's method for estimation of nitrogen present in a soil sample, ammonia evolved from 0.75 g of sample neutralised 10 mL of 1 M H_2SO_4 . The percentage of nitrogen in the soil is (2014)
- 45.33
 - 35.33
 - 43.33
 - 37.33

Answer Key

- | | | | | | | | | | |
|---------|---------|---------|------------|---------|---------|---------|---------|---------|---------|
| 1. (a) | 2. (d) | 3. (b) | 4. (d) | 5. (c) | 6. (d) | 7. (a) | 8. (a) | 9. (b) | 10. (b) |
| 11. (d) | 12. (b) | 13. (a) | 14. (b) | 15. (c) | 16. (a) | 17. (c) | 18. (b) | 19. (b) | 20. (d) |
| 21. (a) | 22. (c) | 23. (d) | 24. (a, b) | 25. (a) | 26. (d) | 27. (b) | 28. (b) | 29. (a) | 30. (b) |
| 31. (c) | 32. (c) | 33. (b) | 34. (d) | 35. (b) | 36. (d) | 37. (a) | 38. (a) | 39. (d) | 40. (a) |
| 41. (d) | 42. (a) | 43. (b) | 44. (d) | | | | | | |